

GeoCarb Keystone/Smile/ Dispersion Bias Study

What's been established, what's still open, and the concrete next steps for reducing retrieval bias — via co-retrieved covariates and new nuisance parameters — once you're back.

As of 2026-07-28 · full detail in `KEYSTONE_SMILE_BIAS_PLAN.md`, cross-referenced by §-number below

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§1

What this project is

GeoCarb's real ground-test calibration report (`keystone_report.pdf`) documents retrieval anomalies — non-convergence, residual striping — tied to keystone/smile geometric distortion and the rectify-then-retrieve L1B pipeline. This project builds a synthetic simulator (`geocarb_gert` , on the `gert` forward-model/retrieval library) that reproduces the *real* GD polynomials, detector geometry, and PSF — then uses it to isolate which mechanisms actually drive bias, at what magnitude, and whether a joint multi-band retrieval (the real GeoCarb L2 approach) helps.

Method throughout: tie each hypothesis to a concrete real-data symptom (§10's evidence table in the full plan), test it in simulation, confirm or rule it out, move to the next candidate.

§2

Established — single-band mechanisms

§9b • confirmed vs. real data

Row-crossing / pixel-grid aliasing — the leading single-band bias mechanism. A detector row averages up to ~10 true along-slit rows near the slit edges. Matches real ground-test non-convergence directly.

§9c • confirmed vs. real data

Clocking is a per-band offset of the keystone curve (FPA2's zero-point sits at row 25 of 1024), not a separate mechanism.

§9d • confirmed vs. real data

FPA2 has an independent calibration-coverage gap (insufficient laser wavelength coverage during ground test), compounding the above.

§9h • simulation only

PSF×smile coupling creates an irreducible, order-independent CO₂ bias (up to ~-5.9 ppm) even on a spatially uniform scene.

§9l/§9m • simulation only — biggest gap

Rectification (regrid-then-retrieve) bias dominates raw geometric distortion — tens-of-ppm, roughly position-independent, ~10–14% outright non-convergence at full row density. The single largest simulated bias source, and not yet checked against real data.

§9j • simulation only

Barcode / scene-transition bias is real but narrowly localized — a sharp spike within ~4–10 rows of a genuine scene boundary, zero elsewhere.

§9x • revised this session

Dispersion order matters enormously for reported χ^2 , independent of fit quality. Originally read as "native fits much better than undistorted" — this session found most of that gap is a numerical artifact, not real fit quality. See §4.

Established — multi-band / joint retrieval

§11b / §11f

Cross-band co-registration is solved. Real slit angle `s` [deg] — never normalized `eta`, which differs per FPA — is the correct shared coordinate, confirmed by the actual ground-test calibration methodology. `nearest_row_pairing_multi()` pairs rows to ≤ 0.5 -row error for any $N \geq 2$ bands, and error doesn't compound as bands are added.

§11g

Joint retrieval measurably helps. FPA0(O2-A)+FPA2(CO2), no aerosol: joint retrieval cuts CO2 std by $\sim 4.4\times$ vs. single-band by breaking the H2O/surface-pressure degeneracy — the real-world rationale for multi-band L2, now demonstrated here.

§11j

The joint pipeline generalizes to any $N \geq 2$ bands (- - fpas 0,1,2,3 -style), smoke-tested at 3 bands.

§11i

Barcode joint-failure mode diagnosed: narrow chi2-dominating spikes hit a wider row fraction jointly, since each band has independent keystone geometry — expected, not a bug.

§11i

Rectified inherits (and compounds) the single-band rectification bias in the joint case too.

This session's headline finding

The dispersion-order convolution artifact

Symptom: the "undistorted" pipeline (zero-distortion baseline, no dispersion correction by original design) showed chi2 and gas-bias scatter far worse than "native" — backwards, since undistorted should be the *cleanest* case.

ROOT CAUSE, TRACED TO SOURCE

gert/forward_model.py only uses exact (sub-grid) ILS-kernel centering when dispersion parameters are present in the state vector (`wn_centers` is not `None` triggers `ILS.convolve's exact_center=True`); absent dispersion it silently snaps each channel center to the nearest hi-res grid point instead. This project's truth-rendering path always uses exact centering. So any retrieval that never floats dispersion is structurally stuck comparing against a mismatched convolution convention — a small, real, near-row-independent chi2 floor with nothing to do with retrieval quality. Confirmed directly: evaluating the forward model at the *true* prior state, no optimizer involved, reproduces the identical mismatch for native and undistorted alike, matched to 4 significant figures.

Fix: float dispersion ($\text{order} \geq 1$) in every pipeline, including "undistorted" — not because it has real calibration error to correct, but purely to trigger the matching exact-center convolution path. Retrieved coefficients are expected to sit near zero. Confirmed via a dedicated diagnostic (`gd_joint_undistorted_dispersion_diag.py`): collapses undistorted's chi2 to native's level or better, across uniform, uniform+noise, and realistic scenes. Applied to both the single-band and multi-band scripts and all five associated plotting scripts. Full evidence: §11k.

OPEN THREAD THIS CREATES

§9x had earlier attributed *some* of native's dispersion-driven chi2 advantage to a real, keystone-correlated wavelength-registration error (small, row-dependent, near-zero at the smile-slope-null row). That may still be a real, smaller effect riding on top of the larger numerical artifact found this session. Disentangling the two is the first concrete next step — see §6.1.

§5

In-flight right now

All jobs below apply the §4 fix and supersede all prior results for the same FPA / band-pair — old results and their old-format plots were already deleted.

SLURM job status as of this writing — check with `squeue -u $USER`

BATTERY	JOB	SCOPE	STATUS
Single-band, all 4 FPAs	1604089–1604090, 1604097–1604100	Full 3-scene × 2-noise × 4-FPA matrix (24 array tasks)	running
FPA0+FPA3 joint	1603981–1603986		

BATTERY	JOB	SCOPE	STATUS
		native + undistorted only (rectified unaffected, not rerun)	running
FPA0+FPA2 joint	1604101–1604106	Full 3 pipelines	queued
FPA0+FPA1 joint	1604107–1604112	Full 3 pipelines	queued

Check with `module load slurm/24.05.0 && squeue -u $USER` (or add `/gpfs/fs1/sfw3/rhel9-x86_64/slurm/24.05.0/bin` to `PATH` directly — `squeue / sbatch` aren't reliably on `PATH` in fresh shells on this cluster, same as `git` needing `module load git/2.50.1`).

Once these land, regenerate plots:

```
python scripts/gd_band_stress_test_plot.py           # per FPA, run 4x
python scripts/gd_band_stress_test_plot_grid.py
python scripts/gd_band_stress_test_plot_residual_spectra.py
python scripts/gd_joint_band_plot.py --fpas 0,1     # and --fpas 0,2 / 0,3
python scripts/gd_joint_band_plot_residual_spectra.py --fpas 0,1  # slow, re-renders
```

§6

Concrete next steps

6.1 Immediate — disentangle real vs. numerical dispersion signal

Cheap, high-value, no new runs needed. Once the single-band FPA1 rerun lands, redo §9x's row-0/512/1023 `disp_a0` comparison table, now that undistorted also floats dispersion. If undistorted's `disp_a0` vs. row now tracks the same small, row-dependent pattern native's already does (near-zero at slit center, growing toward the edges, tracking §9w's row-crossing) — that confirms §9x's real-registration-error story as a smaller effect layered on the artifact §4 fixed. If undistorted's `disp_a0` stays flat everywhere while native's still shows the old pattern, the row-dependence was itself an artifact worth another look.

6.2 Foundational — real-data validation

Almost nothing in §2/§3 has been checked against real data yet — most rows in the full plan's §10 evidence table say "confirmed in simulation, not yet checked against real data." Before

investing heavily in a bias-correction scheme tuned to these simulated mechanisms, worth confirming the mechanisms themselves show up in real ground-test or on-orbit data:

- Does real data's non-convergence rate on rectified/regridded processing actually exceed native-grid processing by the ~10%+ margin simulation predicts (§9m)? Highest-value single check — rectification bias is the largest mechanism found.
- Does real spectral residual show the line-locked structure §9n found for rectification error, vs. the smooth PSF×smile-coupling shape (§9h)? These imply different correction strategies.
- Does keystone-heterogeneity contamination in real scene transitions show the same narrow (~4–10 row) footprint §9j predicts?

If real data doesn't show a predicted signature, that's just as valuable — it re-scopes which mechanism(s) actually matter for correction.

6.3 Bias correction via retrieved covariates

Every retrieval already saves its **full state vector** (`_state_full`: every element's retrieved value, prior, posterior 1σ) and **post-fit residual**, per row — a standing rule for this project specifically so this analysis is always possible without a rerun.

1. **Regress each pipeline's gas bias against its own co-retrieved nuisance parameters** — `T_offset`, `albedo`, `albedo_slope`, and especially the dispersion coefficients, which per §9x/§4 directly track keystone/row-crossing amount. A simple `bias_corrected = bias_raw - f(dispatch_a0, dispatch_a1, dispatch_a2, ...)` fit against the already-existing §9v/9w and §11g–11i results could measurably reduce along-slit bias with no new physics or new retrieval runs — the cheapest path to a result.
2. **Use real slit angle `s` itself** (or distance from each FPA's own keystone-null / smile-slope-null row, already tabulated per FPA in §9i/§11i) as an explicit covariate — a position-aware correction rather than one global bias number, directly testable against the existing along-slit sweep data.

6.4 New nuisance parameters to explore

1. **A spatial/spectral surface-albedo basis** beyond the current constant + linear-slope model — `gert.forward_model`'s `SurfaceBasis` machinery already exists but hasn't been exercised here. Worth checking whether a higher-order albedo basis absorbs any of the rectification-interpolation residual, which §9n found has real spectral structure a linear slope can't represent.
2. **Finish the calibration-mismatch experiment (§12)**. Built and smoke-tested for wavelength-only mismatch; slit-position mismatch needs `wavelength_slit_to_xy_assumed()` + a `rectify_assumed()` counterpart, not yet built. Matters directly for bias correction: §12c found wavelength-calibration error is partially self-correcting (a dispersion nuisance parameter absorbs

it) while slit-position/keystone error has **no equivalent nuisance parameter anywhere in `StateVector.gas_scaling()` today** — structurally uncorrectable under the current design unless a new parameter is added for it.

3. **PSF-area-weighted cross-band co-registration** (§11c) as an alternative to nearest-row pairing — flagged as a possible refinement if nearest-row's ≤ 0.5 -row discretization error turns out to matter in practice; not built, not yet shown necessary.

6.5 Aerosol — deliberately deferred, still open

Every joint-retrieval result in §3 is explicitly aerosol-free (the aerosol–CO₂ degeneracy would compound with whatever's found here, and costs much more compute). `ForwardModel` already has an aerosol Jacobian path ready to use. §11d items 4/6 lay out the plan: extend to a shared atmosphere with an aerosol layer, then compare CO₂-only vs. CO₂+O₂A+aerosol joint retrieval, with a stated hypothesis (aerosol shouldn't touch the rectification-interpolation bias, since that's a spectral-registration artifact independent of photon path length) worth confirming rather than assuming. Any bias-correction scheme from 6.3/6.4 should be re-checked once aerosol is added, since real data has both effects simultaneously.

§7

Where things live

`KEYSTONE_SMILE_BIAS_PLAN.md`

The full chronological record; this summary's source of truth. Search by "§N" section number.

`geocarb_gert/`

`gd_render.py`

(truth rendering),

`gd_polynomials.py`

(real GD polynomials),

`cross_band.py`

(co-registration),

`along_slit_scene.py`

(truth-atmosphere construction).

`scripts/gd_band_stress_test*.py`

Single-band battery + 4 plotting scripts.

`scripts/gd_joint_band_test*.py`

Multi-band battery (any $N \geq 2$ FPAs) + 2 plotting scripts.

```
scripts/gd_joint_undistorted_dispersion_diag.py
```

The diagnostic that found/confirmed §4's fix; not part of the regular battery, kept as a standing record.

```
results/*.pkl
```

Not git-tracked. Regenerate via the battery scripts, never hand-edit.

```
plots/*.png, *.pdf
```

Git-tracked; regenerated by the

```
*_plot*.py
```

scripts reading from

```
results/
```

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Written as a pre-break checkpoint, 2026-07-28. Everything above is cross-referenced to `KEYSTONE_SMILE_BIAS_PLAN.md` — that document is authoritative if anything here goes stale.